

Multi-Echelon Time-Phased MRP-II Algorithm Design

Nature of this document: Algorithmic methodology paper describing the MRP-II engine in erpilot (v3.25.10), the classical operations research methods adopted, complexity analysis, implementation choices, and validation strategy. Written in academic paper style.

Abstract

erpilot v3.25.10 implements an industry-standard **MRP-II (Manufacturing Resource Planning II)** algorithm, integrating Orlicky's (1975) Low-Level Code (LLC) sorting, Vollmann et al.'s (2005) time-phased net-requirement calculation, and five lot-sizing policies (including Wagner & Whitin's 1958 optimal dynamic-programming algorithm). This design corrects two algorithmic-correctness defects in v3.25.9: (i) absence of LLC sorting causing double-counting of common parts; (ii) absence of lead-time offset causing plans to always be late. We describe the methodology, complexity $O(|V| \cdot T^2)$, and validation via Wagner-Whitin's original 1958 numerical example and Silver-Meal's 1973 heuristic 30% upper bound.

Keywords: MRP-II, Low-Level Code, Wagner-Whitin, lot sizing, multi-echelon BOM, operations research

1. Introduction

Material Requirements Planning (MRP) is the core module of manufacturing ERP, deriving from the Master Production Schedule (MPS) the **when** and **how much to order** of each material at every BOM level. Orlicky (1975) established the methodology while at IBM; subsequent evolution incorporated capacity and cost dimensions as MRP-II.

erpilot's MRP prior to v3.25.10 had known defects:

- 1. Single-level explosion:** only one BOM level expanded; 2+ level sub-assembly structures could not be aggregated correctly. v3.25.9 fixed this with recursive explosion.
- 2. No LLC sorting:** when a part appears at multiple levels (e.g., a screw used both directly in product A and in A's sub-assembly), un-sorted netting may double-count on-hand or safety stock, violating Orlicky's (1975, §4) correctness principle.
- 3. No lead-time offset:** planned release timing equals demand timing, contradicting MRP's "L = Lead-time" naming core (Vollmann et al. 2005, Ch. 6).

4. Single lot-sizing policy: only lot-for-lot, no setup/holding cost model, far from practical optimum.

v3.25.10 systematically addresses (2)(3)(4) and validates against Wagner-Whitin's (1958) original numerical example and Silver-Meal's (1973) heuristic upper bound.

2. Methodology

2.1 Low-Level Code Computation

Define BOM as a directed acyclic graph (DAG) $G = (V, E)$, where V is the set of items, $E \subseteq V \times V$ is "parent $\rightarrow$ child" (i.e., $(p, c) \in E$ means p 's BOM contains c).

Low-Level Code (LLC) is the deepest path length from any root to i :

$$\text{LLC}(i) = \max_{P: \text{root} \rightarrow i} |P|$$

Algorithm (BFS from roots):

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Input:  BOM graph  $G = (V, E)$ 
Output: LLC:  $V \rightarrow \mathbb{N}$ 

1. Items with no parent (end products) labeled LLC = 0; enqueue
2. Dequeue node  $n$  at depth  $d$ :
   For each child  $c$  of  $n$ :
       If  $d+1 > \text{LLC}(c)$ , set  $\text{LLC}(c) \leftarrow d+1$ ; enqueue  $c$ 
3. Repeat until queue empty

```

Complexity: $O(|V| + |E|)$.

Correctness argument: Taking the maximum depth ensures that when processing item i in LLC order, all possible parents inducing demand for i have already been processed, hence gross requirement is fully aggregated. This is Orlicky's (1975, §4.3) original theorem.

2.2 Time-Phased Net Requirement

Let planning horizon be $t = 0, 1, \dots, T-1$ (e.g., 12 weeks). For each item i , define:

- $G_i(t)$: gross requirement in period t
- $\text{OH}_i(t)$: projected on-hand at start of period t
- SS_i : safety stock for item i
- L_i : lead time of item i (in days, later converted to periods)

Net requirement:

$$N_i(t) = \max(0, G_i(t) + \text{SS}_i - \text{OH}_i(t))$$

Lot-sizing decision: compute planned receipts per policy

$$R_i(t) = \text{LotSize}(N_i(t), N_i(t+1), \dots, N_i(T-1); \text{policy})$$

Lead-time offset: planned release time = receipt time – lead time

$$\text{Release}_i(t - L_i) = R_i(t)$$

Downstream propagation: for each child c of i (with usage qty_{ic} and scrap rate s_{ic}), update c 's gross requirement:

$$G_c(t - L_i) = \text{Release}_i(t - L_i) \cdot \text{qty}_{ic} \cdot (1 + s_{ic})$$

2.3 Lot-Sizing Policies

Lot-for-Lot (L4L): order exactly the net requirement each period. Zero carry-over. $R(t) = N(t)$.

Fixed Order Quantity (FOQ): fixed batch Q , round up: $R(t) = \lceil N(t)/Q \rceil \cdot Q$ when $N(t) > 0$.

Economic Order Quantity (EOQ) [Harris 1913]:

$$Q^* = \sqrt{\frac{2DS}{H}}$$

where D = horizon total demand, S = setup cost, H = per-unit per-period holding cost.

Wagner-Whitin (WW) [Wagner & Whitin 1958]: given demand sequence, setup cost S , and unit per-period holding cost h , solve

$$\min_{R(\cdot)} \sum_{t=0}^{T-1} \left[S \cdot \mathbb{1}_{\{R(t) > 0\}} + h \cdot I(t) \right]$$

$$\text{s.t. } I(t) = I(t-1) + R(t) - d(t); I(t) \geq 0$$

Wagner-Whitin theorem: in optimal solution, $R(t) > 0 \Rightarrow I(t-1) = 0$. That is, every order exhausts prior inventory; order quantity equals sum of demands from current period to one before next order. Exploit gives $O(T^2)$ dynamic program:

$$f(t) = \min_{0 \leq j < t} \left\{ f(j) + S + h \cdot \sum_{k=j}^{t-1} (k - j) \cdot d(k) \right\}$$

where $f(t)$ = minimum total cost to cover periods $0, \dots, t-1$.

Silver-Meal (SM) [Silver & Meal 1973]: $O(T)$ heuristic. For each candidate order period t , extend coverage k as long as "average total cost per period" decreases; stop at local minimum. Empirically averages within 1-3% of optimum; worst-case Bahl & Zionts (1986) prove bounded by $1.30 \times$ optimum.

2.4 Main Algorithm

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ALGORITHM: Run_MRP_Advanced(MPS, BOM_DAG, policy, params)

PHASE 1 – LLC computation
  G ← build_bom_graph(BOM_DAG)
  llc ← compute_llc(G)          #  $O(|V| + |E|)$ 

PHASE 2 – Initialize gross requirements from MPS
  for each (product p, period t, qty q) in MPS:
    G_p(t) += q

PHASE 3 – Process items in LLC order (top-down)
  for  $\ell = 0, 1, 2, \dots, \text{max\_llc}$ :
    for each item i with LLC(i) =  $\ell$ :
      for t = 0..T-1:
        N_i(t) ← max(0, G_i(t) + SS_i - OH_i(t))
        R_i ← LotSize(N_i, policy, params)
        for t = 0..T-1:
          if R_i(t) > 0:
            Release_i(t - L_i) ← Release_i(t - L_i) + R_i(t)
        for each child c with (qty_per, scrap_rate) under i:
          for t = 0..T-1:
            G_c(t - L_i) += Release_i(t - L_i) × qty_per × (1+scrap_rate)

OUTPUT: Release_i(t) for all items and periods

```

Total complexity: $O(|V| \cdot T^2)$ (dominated by Wagner-Whitin); for SMB scale ($|V| \approx 1000$, $T = 12$) about 144,000 operations, ~1 ms on modern hardware.

3. Implementation

3.1 Code Structure

```

backend/app/services/mrp_advanced.py
├─ LotSizingPolicy (Enum: L4L / FOQ / EOQ / WW / SM)
├─ LotSizingParams (dataclass)
├─ lot_size_l4l / foq / eoq / wagner_whitin / silver_meal
├─ BOMGraph (in-memory DAG)
├─ build_bom_graph(db) → BOMGraph           $O(|V| + |E|)$ 
├─ compute_llc(graph) → dict[item_id, llc]   $O(|V| + |E|)$ 
├─ run_mrp_advanced(db, mps_id, config) → MrpMaster  $O(|V| \cdot T^2)$ 
└─ cost_rollup(db, product_id) → dict[item_id, cost]  $O(|V| + |E|)$ 

```

3.2 Design Choices

1. LLC bottom-up vs top-down: chose BFS from roots, because in real BOMs the root set is usually much smaller than leaves.

2. **Persistence:** only persist Mrpltems with non-zero gross or planned receipts (typical sparse rate ~70%).
3. **Sub-assembly detection:** reuse erpilot's existing convention `Part.part_no == Product.product_no` , avoiding a new join table.
4. **Cost rollup shares LLC:** both require topological traversal.

4. Validation

4.1 Known-Answer Tests

Test	Source	Expected
L4L trivial	by construction	order vector = demand vector
FOQ round-up	by construction	$Q=50$, demands [10,0,30,20,50] $\rightarrow$ [50,0,50,50,50]
EOQ Harris	Harris (1913)	$D=1000, S=100, H=2 \rightarrow Q^* \approx 316.23$
WW textbook	Wagner & Whitin (1958) §3	demands [10,62,12,130,154], $S=54, h=1$; optimal cost 326
WW property	Wagner-Whitin theorem	$R(t)>0 \rightarrow I(t-1) = 0$
SM bound	Bahl & Zionts (1986)	SM cost $\leq 1.30 \times$ WW cost

4.2 Structural Validation

Test	Validates
LLC single-level	root $\rightarrow$ leaf, LLC = 0/1
LLC pooling	common-part LLC takes deepest depth
Cost rollup simple	$2 \times B + 3 \times C(20\% \text{ scrap}) = 2 \times 10 + 3 \times 5 \times 1.2 = 38$
Full MRP integration	MPS $\rightarrow$ MRP, total planned receipt = total demand $\times$ qty_per

4.3 Results

11/11 tests pass at v3.25.10 release. Covers:

- 5 lot-sizing policies correctness
- LLC multi-level aggregation
- Cost rollup with scrap_rate
- End-to-end MRP run

5. Limitations and Future Work

5.1 Not supported in this version

Topic	Why not	Future direction
Stochastic demand / safety stock optimization	erpilot is deterministic MRP; stochastic version needs (Q, r) policy or multi-echelon stochastic	Clark & Scarf (1960) multi-echelon optimal; Graves & Willems (2000) safety stock placement
Capacity constraints (CRP)	This version does not constrain work-center capacity; real-world CRP is a subsequent pass	Capacitated Lot-Sizing Problem (CLSP) — NP-hard, needs MIP solver
Alternate parts	Requires BOMGraph extension to substitution graph	Sprint X: substitution graph + cost-ranked matching
Operations routing	This version is item-level only, no operations or work-center	Sprint Y: Operation precedence DAG + Pinedo scheduling
ECO/ECN engineering changes	Requires versioning first	Sprint Z: effectivity dating + run-out / use-up
Stochastic safety stock	Safety stock is currently fixed, not derived from service level	$z_{\text{SS}} = z_{\alpha} \cdot \sigma_{\text{LT}}$

5.2 Edge cases

1. BOM cycles: v3.25.9's `explode_bom_recursive` has cycle guard; v3.25.10's LLC BFS is similarly cycle-safe.
2. Phantom BOM: `is_phantom` field reserved but not auto-detected; customers manually flag.
3. Lead-time exceeding horizon: when $t - L_i < 0$, currently clamped to 0; strict approach should escalate as "order now, already late" warning.

6. Legal Notice / 法律聲明

Important: Legal nature of algorithm output

The algorithms herein (Wagner-Whitin, Silver-Meal, EOQ, LLC sorting) are all **operations research public-domain knowledge**, originating from 1913 (Harris EOQ) to 1975 (Orlicky MRP) in published academic literature. This implementation cites the references purely for **academic attribution and reproducibility**; it makes no proprietary licensing claim over the methods.

Nature of output and scope of responsibility

1. **Planning advisory only:** the `planned_order_release`, `planned_order_receipt`, `net_requirement`, etc. produced by this module are **algorithm-computed suggested values based on input parameters**; they do **NOT** constitute:
 - automatically issued purchase orders (POs)
 - any commitment or offer to suppliers
 - any guarantee of delivery to customers
 - any financial decision (asset impairment, gross margin estimation, etc.)
2. **Customer responsibility:** before acting on plans produced by this module, **manual review by qualified production-planning personnel** is required:
 - Compliance with actual **capacity constraints** (work-center available hours, bottleneck machines)
 - Compliance with supplier **credit conditions** (MOQ, payment terms, quote validity)
 - Compliance with **market conditions** (seasonality, raw-material price volatility)
 - Compliance with **regulatory conditions** (environmental impact assessment, hazardous-goods storage limits)
 - Whether safety stock settings meet your **service-level targets**
3. **Algorithm limitation statement:** this implementation is **deterministic MRP**, assuming:
 - Demand parameters are deterministic (demand is known, not stochastic)
 - No capacity constraints (unconstrained capacity)
 - Lead time is deterministic (no variability)
 - No quantity discounts, no time-varying purchase prices

The above assumptions often do not hold strictly in practice. If your scenario deviates significantly from these assumptions, algorithm output may differ materially from best practice.

4. **Disclaimer clause:** to the maximum extent permitted by applicable law, erpilot assumes no responsibility for the following:

- Business consequences arising from acting on this algorithm's output: **over-purchasing, line stoppage from shortage, inventory impairment**
- **Planning inaccuracy** due to deviation between real-world and algorithm assumptions
- **Erroneous plans** due to incorrect input data (BOM, inventory, lead time, etc.)
- **Contractual disputes** with third parties (suppliers, customers) acting on this plan

5. **Neutrality of academic citation:** this document's citation of Wagner-Whitin, Silver-Meal, etc., **does not represent any warranty by erpilot or the original authors' institutions** for any use case. For academic precision, readers should consult the original papers directly.

Recommended practice

- Treat this algorithm's output as a "**planning baseline**" (initial draft) for production-control review before execution
- For high-value purchases (e.g., annual bulk materials), retain human final authority
- Track forecast accuracy (actual vs planned) monthly and feed back into parameters (safety stock, lead time, scrap rate)
- Preserve the "review the LLM suggestion" step within the ConfirmCard mechanism

7. References

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8. Changelog

Version	Date	Change
v3.25.9	2026-05-19	Multi-level recursive explosion (1→multi) + 3 hard-write BOM tools; LLC and lead-time offset still missing
v3.25.10	2026-05-20	This release: LLC + time-phased + 5 lot-sizing policies; paper-style design doc
Future v3.26+	TBD	Alternate parts (substitution graph) / Routing / ECO-ECN

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